JEE–MAIN EXAMINATION – APRIL 2026

(HELD ON MONDAY 06th APRIL 2026) **TIME : 9:00 AM TO 12:00 NOON**

PHYSICS

TEST PAPER WITH SOLUTION

SECTION-A

26. The density ρ of a uniform cylinder is determined by measuring its mass m, length *l* and diameter d. The measured values of m, *l* and d are 97.42 ± 0.02 g, 8.35 ± 0.05 mm and 20.20 ± 0.02 mm, respectively. Calculated percentage fractional error in ρ is _____ .

(1) 0.63%

(2) 0.82%

(3) 0.72%

(4) 0.25%

Sol. $m = 97.42 \pm 0.02$ g

$l = 8.35 \pm 0.05$ mm

$d = 20.20 \pm 0.02$ mm

$$\rho = \frac{m}{\frac{\pi}{4}d^2\ell}$$

$$\Rightarrow \frac{\Delta\rho}{\rho} = \frac{\Delta m}{m} + \frac{2\Delta d}{d} + \frac{\Delta\ell}{\ell}$$

$$= \frac{0.02}{97.42} + 2\times\frac{0.02}{20.20} + \frac{0.05}{8.35} = 0.0082$$

% error = 0.82%

27. The potential energy of a particle changes with distance x from a fixed origin as $V = \frac{A\sqrt{x}}{x+B}$, where A and B are constant with appropriate dimensions. The dimension of AB are _____ .

(1) $[M^1 L^{5/2} T^{-2}]$

(2) $[M^{3/2} L^{5/2} T^{-2}]$

(3) $[M^1 L^2 T^{-2}]$

(4) $[M^1 L^{7/2} T^{-2}]$

Sol. $[B] = [x] = L$

$$[U] = \left[\frac{A\sqrt{x}}{B+x}\right] \Rightarrow M L^2 T^{-2} = \frac{[A]L^{1/2}}{L}$$

$[A] = M L^{5/2} T^{-2}$

28. The rain drop of mass 1 g, starts with zero velocity from a height of 1 km. It hits the ground with a speed of 5 m/s. The work done by the unknown resistive force is _____ J. (take $g = 10$ m/s^2)

(1) – 8.75

(2) – 8.35

(3) – 9.55

(4) – 9.98

Sol. $w_g + w_{res} = \Delta k$

$$mgh + w_{res} = \frac{1}{2}mv^2$$

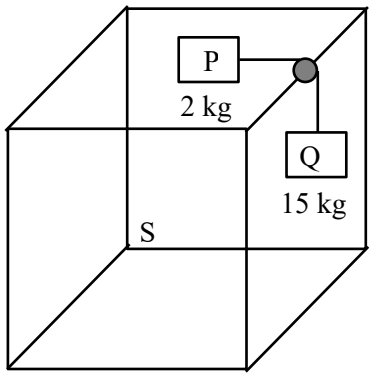
$$\therefore w_{res} = \frac{1}{2}mv^2 - mgh$$

$$= 10^{-3}\left(\frac{25}{2} - 10^4\right)$$

$= -9.98$ J

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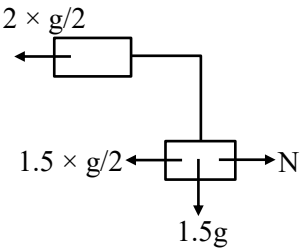
29. Two blocks (P and Q) with respectively masses 2 kg and 1.5 kg are joined by a massless thread. These blocks are mounted on a frictionless pully which is fixed on the edge of a cube (S), as shown in the figure below. Block P is positioned on the top surface which has no friction and block Q is in contact with side-surface, having coefficient friction μ. The cube (S) moves towards the right with acceleration of $\frac{g}{2}$, where g is gravitational acceleration. During this movement the block P and Q remain stationary. The value of μ is _____ . (take g = 10 m/s^2)



(1) 0.33 (2) 0.67

(3) 1 (4) 0.5

Sol. w.r.t bigger block



$$f = \mu N = \mu \times \frac{1.5g}{2}$$

$$1.5g = g + \mu + \frac{1.5g}{2} \quad \Rightarrow 0.5 = \mu \times \frac{1.5}{2}$$

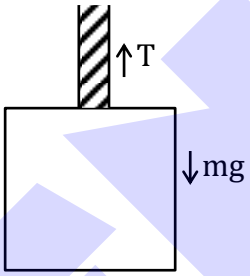
$$\left(\mu = \frac{2}{3}\right)$$

30. A lift of mass 1600 kg is supported by thick iron wire. If the maximum stress which the wire can withstand is 4×10^8 N/m^2 and its radius is 4 mm, then maximum acceleration the lift can take is _____ m/s^2. (take g = 10 m/s^2 and π = 3.14)

(1) 2.56 (2) 3.89

(3) 4.32 (4) 5.16

Sol.



$T - mg = ma$

$(\sigma\pi r^2) - mg = ma$

$\Rightarrow 4 \times 10^8\ \pi\ (4 \times 10^{-2})^2 - 16000 = 1600\ a$

$\Rightarrow\ a = 2.56$ m/s^2

31. A solid sphere of radius 4 cm and mass 5 kg is rotating (rotation axis is passing through the centre of the sphere) with an angular velocity of 1200 rpm. It is brought to rest in 10 s by applying a constant torque. The torque applied and the number of rotations it made before it comes to rest are _____ and _____ respectively.

(1) 0.128 π Nm, 100 (2) 0.0128 π Nm, 50

(3) 0.128 π Nm, 50 (4) 0.0128 π Nm, 100

Sol. $\omega_i = 1200 \text{ rpm} = 1200 \times \frac{2\pi}{60} = 40\ \pi \text{ rad /sec}$

$\rightarrow \omega_f = \omega_i + \alpha t \Rightarrow 0 = 40p + \alpha(10)$

$\alpha = -4\pi$ rad/sec^2

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$\rightarrow I = \frac{2}{5}MR^2 = \frac{2}{5}(5)(0.04)^2 = 2 \times 0.0016$

$I = 0.0032$ Kg-m^2

$\rightarrow \tau = I\alpha = 0.0032 \times 4\pi$

$\tau = 0.0128\pi$ N-m

$\rightarrow \omega_f^2 = \omega_i^2 + 2\alpha\theta$

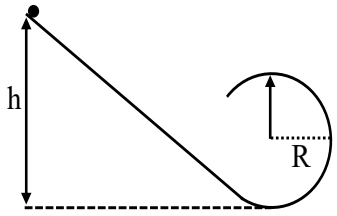
$0 = (40\pi)^2 + 2(4\pi)\theta$

$\theta = \frac{1600\pi^2}{8\pi} = 200\pi$

No. of revolution $= \frac{\theta}{2\pi} = \frac{200\pi}{2\pi}$

No. of revolution = 100

32. A smooth inclined plane ends in a vertical circular loop, as shown in the figure. A small body is released from height h as shown. If the body exerts a force of three times its weight on the plane at the highest point of circle then the height h = αR. The value of α is



(1) 2 (2) 4

(3) 3 (4) 6

Sol. at highest point

$mg + N = \frac{mv^2}{R}$

$4\,mg = \frac{mv^2}{R}$

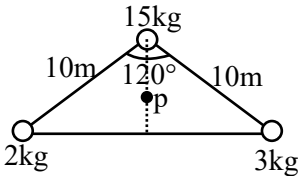
$v = \sqrt{4gR}$

Now energy conservation

$mg(h - 2R) = \frac{1}{2}m \times 4gR$

$(h = 4R)$

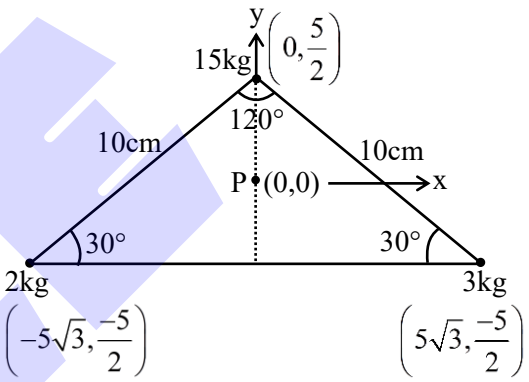
33. The position of center of mass of three masses 2 kg, 3 kg and 15 kg placed with respect to mid point (p) of normal bisector, as shown in the figure is ____ .



(1) $\left(\frac{\sqrt{3}}{4}, 1.25\right)$ (2) $\left(\frac{\sqrt{3}}{4}, 1.0\right)$

(3) (0, 0) (4) (1.25, 0)

Sol.



$X_{cm} = \frac{2 \times (-5\sqrt{3}) + 3(5\sqrt{3}) + 15 \times 0}{20} = \frac{\sqrt{3}}{4}$ cm

$Y_{cm} = \frac{2 \times (-5/2) + 3(5/2) + 15 \times (5/2)}{20}$

$Y_{cm} = 1.25$ cm

34. The two wires A and B of equal cross-section but of different materials are joined together. The ratio of Young's modulus of wire A and wire B is 20/11. When the joined wire is kept under certain tension the elongations in the wires A and B are equal. If the length of wire A is 2.2 m, then the length of wire B is____m.

(1) 1.1 (2) 2.22

(3) 1.21 (4) 4.44

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Sol. $\Delta\ell = \frac{T\ell}{YA}$

$\ell \propto Y$

$\frac{\ell_1}{\ell_2} = \frac{Y_1}{Y_2} = \frac{20}{11}$

$\because \ell_1 = 2.2\,m$

$\therefore \ell_1 = 1.21\,m$

35. Two closed vessels of same volume are joined through a narrow tube and both vessels are filled with air of pressure 90 kPa and temperature 400 K. Keeping the temperature of one vessel constant at 400 K the second vessel temperature is raised to 500 K. The final pressure in the vessels is _____ kPa.

(1) 100 (2) 120

(3) 90 (4) 105

Sol. Using mole conservation

$\frac{2\times9\times10^4\times V}{R\times400} = \frac{P\times V}{R\times400} + \frac{P\times V}{R\times500}$

P = 100 kPa

36. In interference experiment the path difference between two interfering waves at a point A on the screen is λ/3, where λ is the wavelength of these waves, and at another point B the path difference is λ/6. The ratio of intensities at points A and B is _____ .

(1) 3 (2) 4

(3) 1/3 (4) 1/4

Sol. $\Delta\phi_A = \frac{2\pi}{\lambda}\times\frac{\lambda}{3} = \frac{2\pi}{3}$

$\therefore I_A = I + I + 2\sqrt{I}.\sqrt{I}.\cos\left(\frac{2\pi}{3}\right) = I$

$\Delta\phi_B = \frac{2\pi}{\lambda}\times\frac{\lambda}{6} = \frac{\pi}{3}$

$\therefore I_B = I + I + 2\sqrt{I}.\sqrt{I}.\cos\left(\frac{\pi}{3}\right) = 3I$

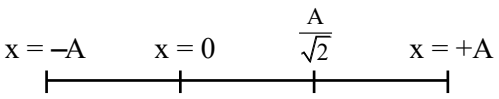
$\frac{I_A}{I_B} = \frac{I}{3I} = \frac{1}{3}$

37. A particle is executing simple harmonic motion. Its amplitude is A and time period is 5 sec. the time required by it to move from x = A to $x = \frac{A}{\sqrt{2}}$ is____sec.

(1) 1/4 (2) 5/4

(3) 5/8 (4) 3/8

Sol.



$x = 0 \rightarrow \theta = 0°$

$x = \frac{A}{\sqrt{2}} \rightarrow \theta = \frac{\pi}{4}$

$\therefore$ phase covered $= \frac{\pi}{4}$

$\text{time} = \frac{\text{phase}}{w} = \frac{\pi}{4w} = \frac{\pi}{4}\times\frac{T}{2\pi} = \frac{T}{8}$

$\therefore t = \frac{5}{8}$ secs.

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38. A thin half ring of radius 35 cm is uniformly charged with a total charge of Q coulomb. If the magnitude of the electric field at centre of the half ring is 100 V/m, then the value of Q is _____ nC.

$(\epsilon_0 = 8.85 \times 10^{-12}\ C^2/Nm^2$ and $\pi = 3.14)$

(1) 2.14 (2) 2.44

(3) 3.25 (4) 0.7

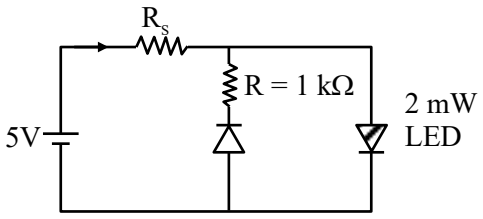
Sol. $E = \frac{2k\lambda}{R}\sin\left(\frac{\theta}{2}\right)$

$100 = 2 \times \frac{9\times10^9 \times Q}{\pi R^2}$

$Q = 2.14 \times 10^{-9}$ C

39. The maximum rated power of the LED is 2 mW and it is used in the circuit with input voltage of 5 V as shown in the figure below. The current through resistance R_S is 0.5 mA.

The minimum value of the resistance of R_S, to ensure that the LED is not damaged is _____ kΩ.



(1) 6 (2) 2

(3) 4 (4) 5

Sol. Diode in reverse bias so no current in it.

$P_{bulb} = 2 \times 10^{-3} = V_{bulb}$

$v_{bulb} = 4V$

So $5V = V_{RS} + V_{bulb}$

$V_{rs} = 5 - 4 = 1$

$1 = 0.5 \times 10^{-3}\ R_s \Rightarrow R_s = 2k\Omega$

40. A point light source emits E.M. waves in free space. A detector, placed at a distance of L m, measures the intensity as I_0. The detector is now shifted to another location on the same spherical surface ensuring the angle between original location and new location as 45°. The measured intensity at new location will be _____.

(1) $\frac{I_0}{4}$ (2) I_0

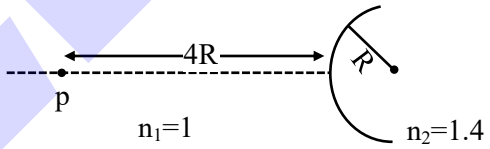
(3) $\frac{I_0}{\sqrt{2}}$ (4) $\frac{I_0}{2}$

Sol. Distance is same

So intensity is also same

$I = \frac{P}{4\pi R^2}$

41. A spherical interface lens of radius R separates two media of refractive indices 1 and 1.4 respectively as shown in the figure below. A point source is placed at a distance of 4R in front of spherical interface. The magnitude of the magnification of point source image is _____.



(1) 1.66 (2) 2.33

(3) 2.66 (4) 1.33

Sol. $m = \frac{n_1}{n_2}\frac{v}{u}$

$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$

$\frac{1.4}{v} + \frac{1}{4R} = \frac{1.4-1}{R}$

$\frac{1.4}{v} = \frac{4}{10R} - \frac{1}{4R}$

$v = \frac{56R}{6} = +\frac{28}{3}R$

$m = \frac{1}{1.4} \cdot \frac{28R/3}{(-4R)} = -1.67$

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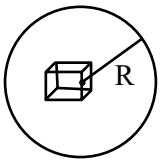
42. A small cube of side 1 mm is placed at the centre of a circular loop of radius 10 cm carrying a current of 2 A. The magnetic energy stored inside the cube is $\alpha \times 10^{-14}$ J. The value of α is _____.

($\mu_0 = 4\pi \times 10^{-7}$ Tm/A, $\pi = 3.14$)

(1) 6.28 (2) 6.28×10^{-6}

(3) 628 (4) 6.28×10^{-4}

Sol. Magnetic field at centre $B = \left(\frac{\mu_0}{4\pi}\right)\frac{2\pi i}{R}$



$$= \frac{10^{-7} \times 2 \times \pi \times 2}{10 \times 10^{-2}}$$

So energy stored

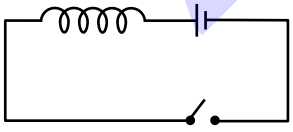
$$U = \frac{B^2}{2\mu_0} \times V$$

$$= \frac{10^{-14} \times 4 \times \pi^2 \times 4 \times 100}{2 \times 4\pi \times 10^{-7}} \times \left(1 \times 10^{-9}\right)$$

$$= 2\pi \times 10^{-14} \text{ J}$$

$= 6.28 \times 10^{-14}$ J

43. An inductor of inductance 10 mH having resistance of 100 Ω is connected to battery of E.M.F. 1.0 V through a switch as shown in the figure below. After switch is closed, the ratio of instantaneous voltages across the inductor when the current passing through it is 2 mA and 4 mA is _____.



(1) 4/3 (2) 3/4

(3) 5/3 (4) 3/5

Sol. Apply KVL

$E - iR - V_L = 0$

$V_L = E - iR = 1 - 100\, i$

at $i_1 = 2 \times 10^{-3}$ A

$V_{L_1} = 0.8$V

at $i_2 = 4 \times 10^{-3}$ A

$V_{L_2} = 0.6$V

$$\therefore \frac{V_{L_1}}{V_{L_2}} = \frac{4}{3}$$

44. The ratio of momentum of the photons of the 1st and 2nd line of Balmer series of Hydrogen atoms is α/β. The possible values of α and β are :-

(1) 27 and 20 (2) 3 and 16

(3) 5 and 36 (4) 20 and 27

Sol. $p = \frac{h}{\lambda}$

$$\frac{1}{\lambda_1} = R\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = R\left(\frac{1}{4} - \frac{1}{9}\right)$$

$$\frac{1}{\lambda_1} = R\frac{5}{36}$$

$$\frac{1}{\lambda_2} = R\left(\frac{1}{2^2} - \frac{1}{4^2}\right) = R\left(\frac{1}{4} - \frac{1}{16}\right)$$

$$\frac{1}{\lambda_2} = \frac{3R}{16}$$

$$\frac{p_1}{p_2} = \frac{\lambda_2}{\lambda_1} = \frac{\frac{16}{3R}}{\frac{36}{5R}} = \frac{20}{27}$$

$\alpha = 20$ & $\beta = 27$

45. A LCR series circuit driven with E_{rms} = 90 V at frequency f_d = 30 Hz has resistance R = 80 Ω, an inductance with inductive reactance X_L = 20.0 Ω and capacitance with capacitive reactance X_C = 80.0 Ω. The power factor of the circuit is _____.

(1) 0.8 (2) 0.64

(3) 0.9 (4) 0.5

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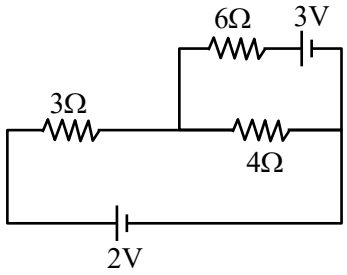
Sol. Power factor is $\cos\phi = \frac{R}{Z}$

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = 100\Omega$$

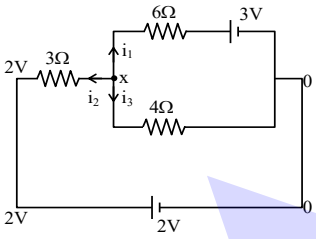
$$\cos\phi = \frac{80}{100} = 0.8$$

SECTION-B

46. Refer to the circuit diagram given below. The heat generated across the 6 Ω resistance in 100 second is $\frac{\alpha}{100}$ J. The value of α is _____. (Nearest integer)



Sol.



$$i_1 + i_2 + i_3 = 0$$

$$\frac{x-0-3}{6} + \frac{x-2}{3} + \frac{x-0}{4} = 0$$

$$x = \frac{14}{9}\text{V}$$

$$\text{Current across } 6\Omega = \frac{3-(14/9)}{6} = \frac{13}{54}\text{A}$$

$$H = \left(\frac{13}{54}\right)^2 \times 6 \times 100 = 34.77 = \frac{\alpha}{100}$$

$$\alpha = 3477$$

47. An unpolarized light of intensity I_0 passes through polarizer and then through a certain optically active solution and finally it goes to analyser. If the angle between analyser and polariser is 0° and intensity of light emerged from analyser is $\frac{3}{8}I_0$, the angle of rotation of the light by the solution with respect to analyser is _____ degrees.

Sol. $\underset{\text{(Unpolarised)}}{I_0} \rightarrow \underset{\text{(Polarised)}}{\frac{I_0}{2}} \xrightarrow{\text{Analyser}}$

$I = I' \cos^2\theta$

$$\frac{3I_0}{8} = \frac{I_0}{2}\cos^2\theta$$

$$\cos\theta = \frac{\sqrt{3}}{2}$$

$$\theta = 30°$$

48. The energy released when $\frac{7}{17.13}$ kg of $^7_3$Li is converted into $^4_2$He by proton bombardment is α × 10^{32} eV. The value of α is _____. (Nearest integer)

(Mass of $^7_3$Li = 7.0183 u, mass of $^4_2$He = 4.004 u, mass of proton = 1.008 u and 1 u = 931 MeV/c^2 and Avogadro number = 6.0 × 10^{23})

Sol. $^7_3\text{Li} + {}^1_1\text{P} \rightarrow 2\,{}^4_2\text{He}$

Δm = (7.018 + 1.008) – 2(4.004) = 0.0183 u

Energy released per reaction E = Δmc^2

$$\text{Number of moles of Li} = \frac{7}{17.13} \times \frac{10^3}{7} = \frac{10^3}{17.13}$$

$$\text{Number of molecules} = \frac{10^3}{17.13} \times 6 \times 10^{23}$$

Total energy released

$$E_{\text{Total}} = 0.0183 \times 931 \times \frac{6}{17.13} \times 10^{32}$$

$$\simeq 6 \times 10^{32}$$

$$\alpha = 6$$

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49. A three coulomb charge moves from the point (0, –2, –5) to the point (5, 1, 2) in an electric field expressed as $\vec{E} = 2x\hat{i} + 3y^2\hat{j} + 4\hat{k}$ N/C. The work done in moving the charge is _____ J.

Sol. $\Delta U = -w_{electrostatic}$

$w_{electrostatic} = -q\Delta V$

$\Delta V = -\int 2xdx - \int 3y^2dy - \int 4dz$

$= \left(-x^2 - y^3 - 4z\right)_{(0,-2,-5)}^{(5,1,2)}$

$= -62$

$w_{electrostatic} = 3(62) = 186$ J

50. A certain gas is isothermally compressed to $\left(\frac{1}{3}\right)^{rd}$ of its initial volume (V_0 = 3 litre) by applying required pressure. If the bulk modulus of the gas is 3×10^5 N/m^2, the magnitude of work done on the gas is _____ J.

Sol. For isothermal process initially B = P_0.

$W = nRT\ell n\left(\frac{V_2}{V_1}\right)$

$= P_0V_0\ell n\left(\frac{1}{3}\right)$

$W = BV_0\ell n\left(\frac{1}{3}\right)$

$W = -988.75$ J ≈ 989 J

Note : Bulk moldulus of gas for isothermal process is not constant. In the given solution we have considered it as initial bulk modulus.

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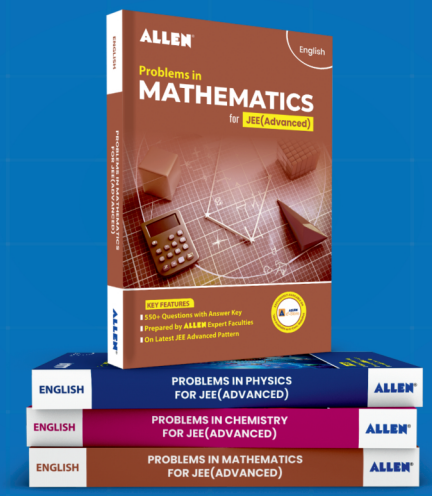


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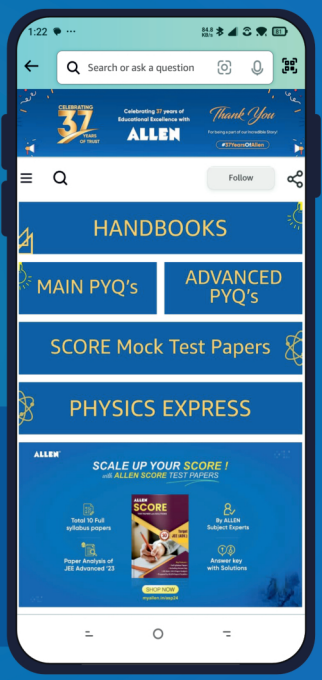
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